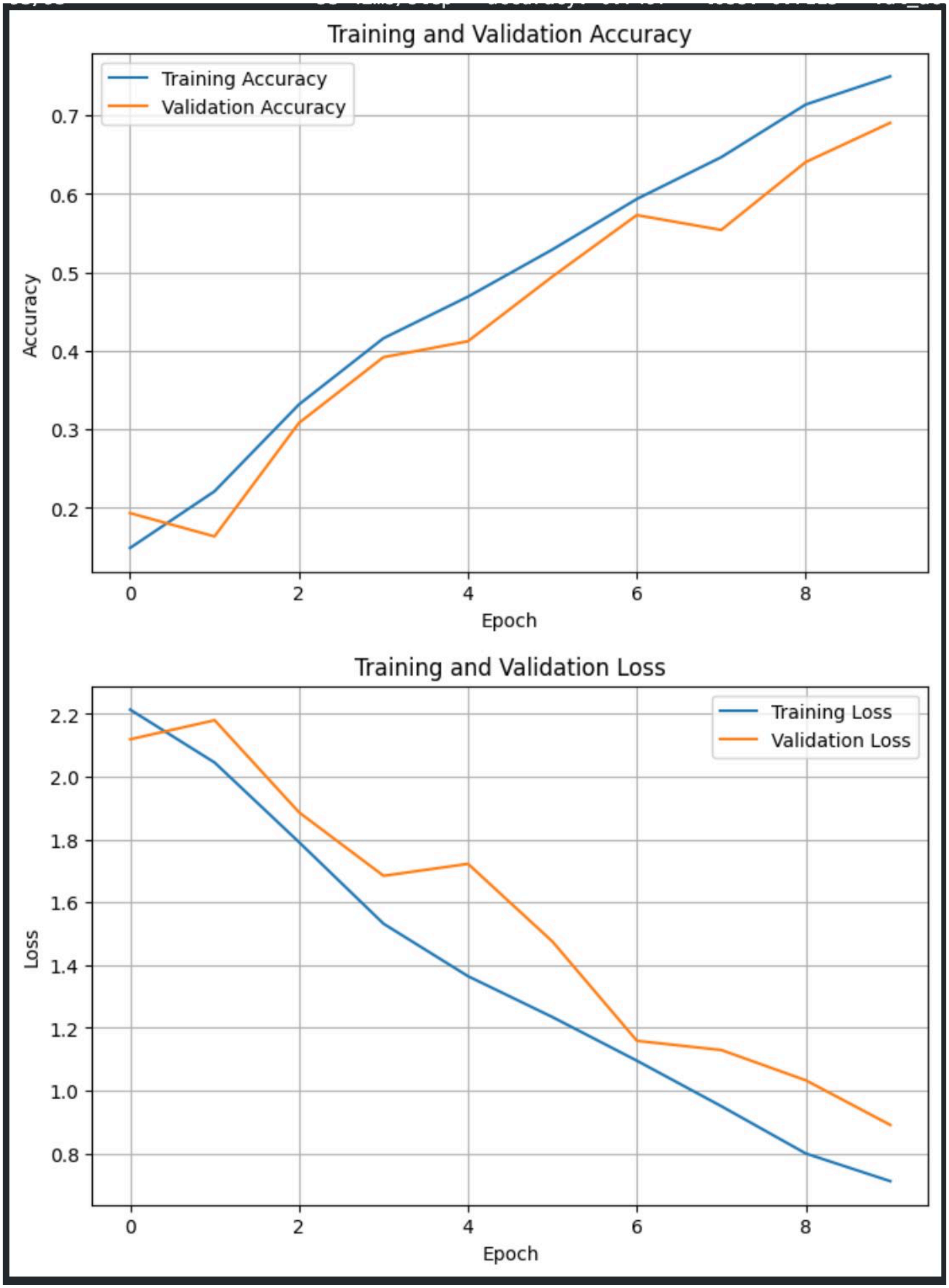
```
========================================================================
CNN-BASED TRAFFIC SIGN RECOGNITION SYSTEM
========================================================================

Parameters
----------------------------------------
Image Size       : 32 x 32
Batch Size       : 32
Epochs           : 10
Images/Class     : 500
Number of Classes: 10

Downloading GTSRB dataset...
Please wait...
Dataset downloaded successfully.
Dataset extracted successfully.
Training directory:
/content/gtsrb_raw/GTSRB/Final_Training/Images

Selected Traffic Sign Classes:
['00000', '00001', '00002', '00003', '00004', '00005', '00006', '00007', '00008', '00009']

Preparing dataset...
Dataset preparation completed.
Found 2964 images belonging to 10 classes.
Found 740 images belonging to 10 classes.
Found 926 images belonging to 10 classes.

Dataset Information
----------------------------------------
Number of Classes : 10
Training Images   : 2964
Validation Images : 740
Testing Images    : 926
```

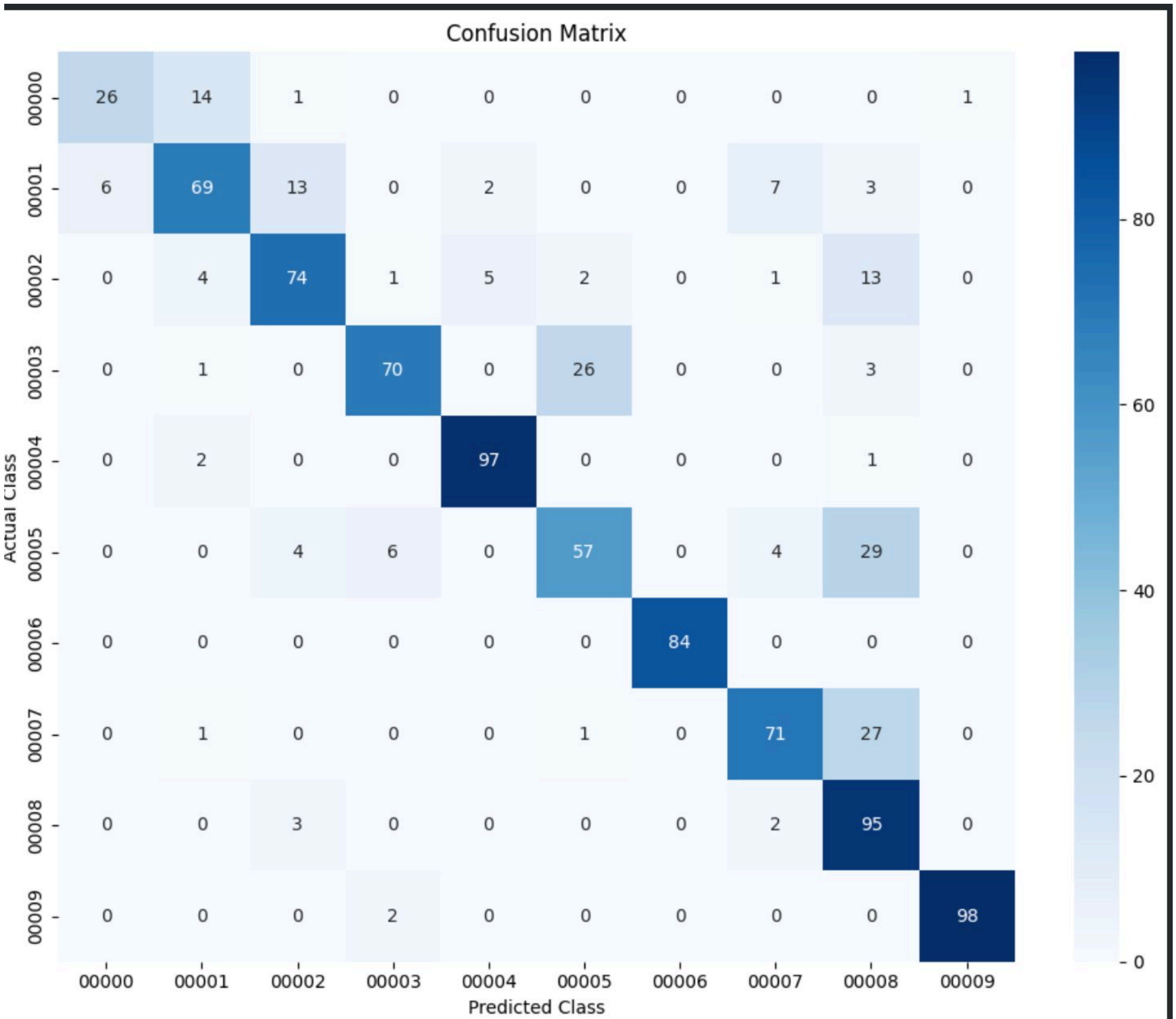

```
CNN MODEL SUMMARY
========================================================================
/usr/local/lib/python3.13/dist-packages/keras/src/layers/convolutional/base_conv.py:113: UserWarning: Do not pass an `input_shape`/`input_dim` argument to a layer. When using Sequential models, prefer using an `Input
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
Model: "sequential"
```

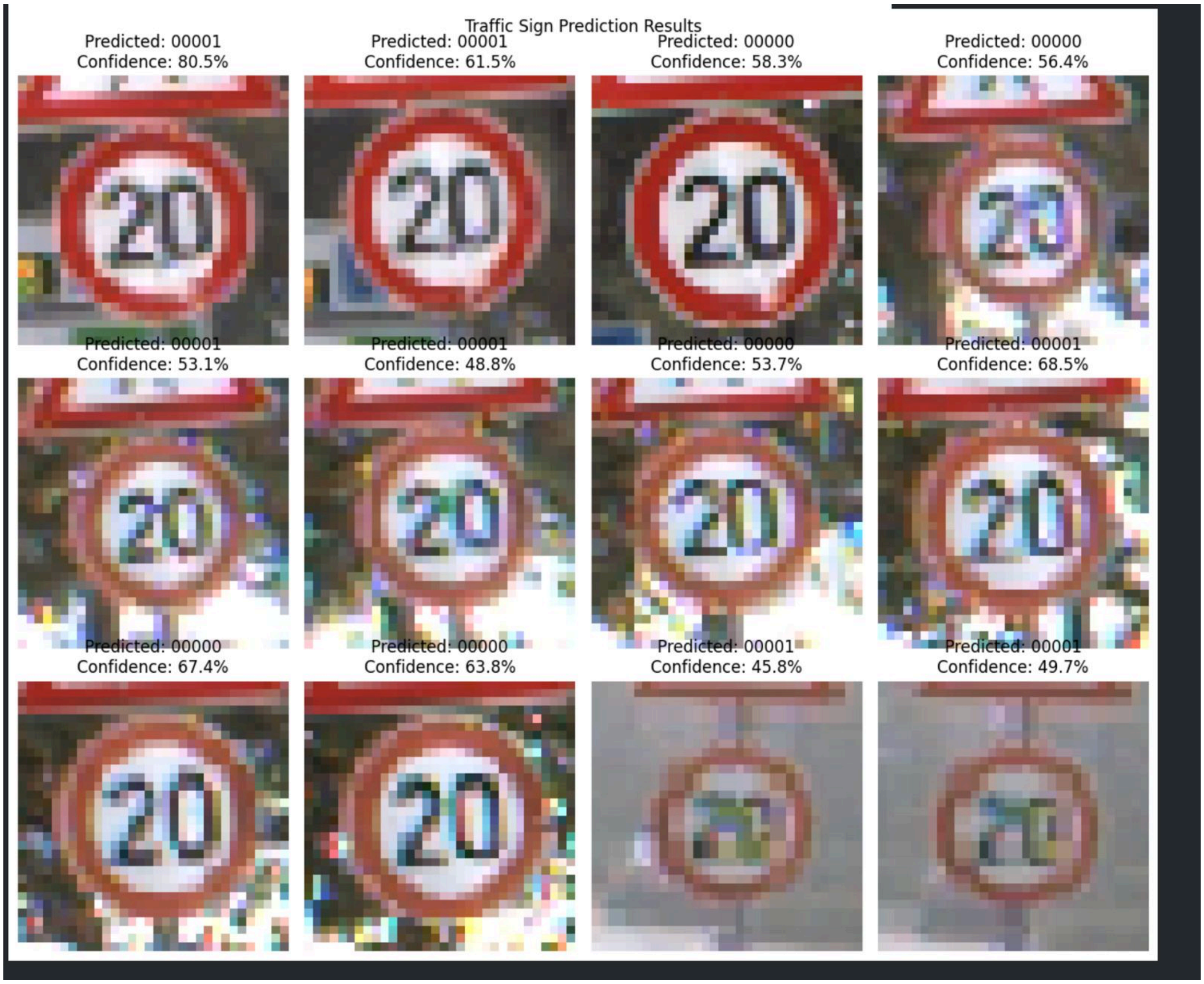
Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 30, 30, 32)	896
max_pooling2d (MaxPooling2D)	(None, 15, 15, 32)	0
conv2d_1 (Conv2D)	(None, 13, 13, 64)	18,496
max_pooling2d_1 (MaxPooling2D)	(None, 6, 6, 64)	0
conv2d_2 (Conv2D)	(None, 4, 4, 128)	73,856
max_pooling2d_2 (MaxPooling2D)	(None, 2, 2, 128)	0
flatten (Flatten)	(None, 512)	0
dense (Dense)	(None, 128)	65,664
dropout (Dropout)	(None, 128)	0
dense_1 (Dense)	(None, 10)	1,290

```
 Total params: 160,202 (625.79 KB)
 Trainable params: 160,202 (625.79 KB)
 Non-trainable params: 0 (0.00 B)

Model compiled successfully.

TRAINING CNN MODEL
========================================================================
Epoch 1/10
93/93 ━━━━━━━━━━━━━━━━━━━━ 6s 53ms/step - accuracy: 0.1488 - loss: 2.2136 - val_accuracy: 0.1932 - val_loss: 2.1192
Epoch 2/10
93/93 ━━━━━━━━━━━━━━━━━━━━ 4s 44ms/step - accuracy: 0.2210 - loss: 2.0451 - val_accuracy: 0.1635 - val_loss: 2.1799
Epoch 3/10
93/93 ━━━━━━━━━━━━━━━━━━━━ 5s 42ms/step - accuracy: 0.3316 - loss: 1.7913 - val_accuracy: 0.3081 - val_loss: 1.8863
Epoch 4/10
93/93 ━━━━━━━━━━━━━━━━━━━━ 6s 60ms/step - accuracy: 0.4160 - loss: 1.5317 - val_accuracy: 0.3919 - val_loss: 1.6843
Epoch 5/10
93/93 ━━━━━━━━━━━━━━━━━━━━ 4s 42ms/step - accuracy: 0.4690 - loss: 1.3645 - val_accuracy: 0.4122 - val_loss: 1.7224
Epoch 6/10
93/93 ━━━━━━━━━━━━━━━━━━━━ 4s 42ms/step - accuracy: 0.5290 - loss: 1.2343 - val_accuracy: 0.4946 - val_loss: 1.4749
Epoch 7/10
93/93 ━━━━━━━━━━━━━━━━━━━━ 5s 52ms/step - accuracy: 0.5935 - loss: 1.0953 - val_accuracy: 0.5730 - val_loss: 1.1587
Epoch 8/10
93/93 ━━━━━━━━━━━━━━━━━━━━ 4s 42ms/step - accuracy: 0.6468 - loss: 0.9503 - val_accuracy: 0.5541 - val_loss: 1.1293
Epoch 9/10
93/93 ━━━━━━━━━━━━━━━━━━━━ 4s 46ms/step - accuracy: 0.7139 - loss: 0.7997 - val_accuracy: 0.6405 - val_loss: 1.0329
Epoch 10/10
93/93 ━━━━━━━━━━━━━━━━━━━━ 5s 42ms/step - accuracy: 0.7497 - loss: 0.7115 - val_accuracy: 0.6905 - val_loss: 0.8907
```


Training and Validation Accuracy
Training Accuracy
Validation Accuracy
Accuracy
Epoch
Training and Validation Loss
Training Loss
Validation Loss
Loss
Epoch

```
TESTING MODEL
======================================================================
29/29 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 0.8002 - loss: 0.5739
29/29 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step


======================================================================
CLASSIFICATION PERFORMANCE
======================================================================
Test Accuracy : 80.02 %
Precision     : 82.12 %
Recall        : 79.29 %
F1-Score      : 79.84 %
Number of Classes: 10
Model Parameters: 160202


======================================================================
CLASSIFICATION REPORT
======================================================================
              precision    recall  f1-score   support

       00000       0.81      0.62      0.70        42
       00001       0.76      0.69      0.72       100
       00002       0.78      0.74      0.76       100
       00003       0.89      0.70      0.78       100
       00004       0.93      0.97      0.95       100
       00005       0.66      0.57      0.61       100
       00006       1.00      1.00      1.00        84
       00007       0.84      0.71      0.77       100
       00008       0.56      0.95      0.70       100
       00009       0.99      0.98      0.98       100

    accuracy                           0.80       926
   macro avg       0.82      0.79      0.80       926
weighted avg       0.82      0.80      0.80       926
```


Confusion Matrix
Actual Class
Predicted Class
00000 00001 00002 00003 00004 00005 00006 00007 00008 00009
26 14 1 0 0 0 0 0 0 1
6 69 13 0 2 0 0 7 3 0
0 4 74 1 5 2 0 1 13 0
0 1 0 70 0 26 0 0 3 0
0 2 0 0 97 0 0 0 1 0
0 0 4 6 0 57 0 4 29 0
0 0 0 0 0 0 84 0 0 0
0 1 0 0 0 1 0 71 27 0
0 0 3 0 0 0 0 2 95 0
0 0 0 2 0 0 0 0 0 98
80
60
40
20
0


Traffic Sign Prediction Results
Predicted: 00001
Confidence: 80.5%
Predicted: 00001
Confidence: 61.5%
Predicted: 00000
Confidence: 58.3%
Predicted: 00000
Confidence: 56.4%
Predicted: 00001
Confidence: 53.1%
Predicted: 00001
Confidence: 48.8%
Predicted: 00000
Confidence: 53.7%
Predicted: 00001
Confidence: 68.5%
Predicted: 00000
Confidence: 67.4%
Predicted: 00000
Confidence: 63.8%
Predicted: 00001
Confidence: 45.8%
Predicted: 00001
Confidence: 49.7%

```
======================================================================
MODEL SAVED SUCCESSFULLY
======================================================================
Model Path: /content/traffic_sign_cnn_model.keras


======================================================================
FINAL RESULTS
======================================================================
Dataset              : GTSRB
Number of Classes    : 10
Input Image Size     : 32 x 32
Training Images      : 2964
Validation Images    : 740
Testing Images       : 926
Test Accuracy        : 80.02 %
Precision            : 82.12 %
Recall               : 79.29 %
F1-Score             : 79.84 %
Model Parameters     : 160202
======================================================================
CNN TRAFFIC SIGN RECOGNITION COMPLETED
======================================================================
```